AI-Assisted Software Migration Architecture

A Dimension-Driven, Traceability-First Framework for Migrating Software Systems Across Technology Stacks

Version: 0.9 - Draft

Abstract

This document describes a reproducible architecture for large-scale, AI-assisted software migration. The framework is technology-agnostic: the same pipeline applies to language migrations, framework shifts, monolith-to-microservices decompositions, data pipeline rewrites, protocol library ports, and modernisations in which a single source repository becomes many target repositories (or many sources consolidate into one).

The framework rests on one foundational concept and three operational mechanisms.

Foundational concept — Dimensional Analysis.
A *dimension* is a structural axis along which source elements are related (dependency graph, data-model graph, service graph, event flow, protocol state machine, security surface, and so on). Dimensions are **discovered per project**, not prescribed. They are the substrate on which the mechanisms below operate: they drive knowledge extraction in Phase 1, partition work into migration phases in Phase 3, and define the per-axis correspondence checks performed during coverage and semantic verification. See the Phase 1 section for how dimensions are inferred and the full catalogue in Appendix A.

Supporting artifact — Platform-Divergence Catalogue.
Alongside the dimension specifications, Phase 1 produces a project-specific catalogue of platform-specific behavioural divergences mined from deep research on the source and target technologies (for example, «source-platform falsy-value semantics differ from target-platform», «source-platform implicit transaction nesting is not portable», «source-platform HTTP header handling requires explicit retention in target»). The catalogue grows through every later phase as new divergences surface during semantic verification, and it feeds back into Gap Resolution and audit checklists. See Appendix F for a representative catalogue from one end-to-end reference project.

Three operational mechanisms.

- **Traceability** — every meaningful target element carries a link back to the source element(s) that shaped it *and* to the specification(s) and decision(s) under which it was produced.
- **Coverage Validation** — programmatic comparison of the source inventory against the target traceability graph, producing a categorical status (covered / eliminated / unmatched) for every source element, broken down *per dimension*.
- **Gap Resolution** — a review cycle in which unmatched source elements are interpreted by an AI agent, placed into the target architecture, and either generate new target code or enhance existing target code. The same cycle rewrites or corrects previously generated target code

when coverage analysis exposes behavioural or structural defects in it.

Together these mechanisms turn AI-assisted migration from a probabilistic, best-effort activity into an auditable, measurable process with an explicit and verifiable notion of *completeness*.

For a catalogue of artifacts produced by an end-to-end application of the framework, see Appendix B.

Pipeline Overview

The framework is a five-phase pipeline. Phases 1–3 run once at project inception (though they may be re-entered iteratively as evidence accumulates). Phase 4 executes repeatedly — one migration phase per working session. Phase 5 begins once enough of the target is ready to run alongside the source.

```
+-------------------------------------------------------------------+
| PHASE 1. Knowledge Extraction                                     |
|   Deep analysis of the source system across all relevant          |
|   dimensions; mining of platform-divergence patterns.         |
|   -> Source architecture specifications (one per dimension)       |
|   -> Requirements document (Mission / Goals / Premises /          |
|      Constraints) + Requirements Traceability Matrix              |
|   -> Platform-divergence catalogue (platform-specific behavioural          |
|      divergences discovered during research; grows through Phases 2-4) |
+-------------------------------------------------------------------+
                            v
+-------------------------------------------------------------------+
| PHASE 2. Decisions                                                |
|   Propose target-architecture and migration-strategy options;     |
|   accept each decision with product team, project architect, SMEs. |
|   -> Decision records (priority-ordered; acceptance may be        |
|      up-front, just-in-time, or parallel with implementation)     |
+-------------------------------------------------------------------+
                            v
+-------------------------------------------------------------------+
| PHASE 3. Target Specifications                                    |
|   ONE unified set of specs covering both the target architecture  |
|   AND migration strategy (order, partitioning, exit gates).       |
|   Generated iteratively from accepted decisions; specs for later  |
|   migration phases may defer on decisions not yet needed.         |
|   -> Migration-phase specifications (spec + plan + tasks)         |
+-------------------------------------------------------------------+
                            v
+-------------------------------------------------------------------+
| PHASE 4. Migration Execution (per-migration-phase loop)           |
|   Traceability + Coverage Validation + Gap Resolution +           |
|   Semantic verification + Integration / End-to-end testing.       |
|   Loops once per migration phase; each run produces target        |
|   artifacts, reports, and a handoff note.                         |
|   -> Target artifacts (code, schema, IaC, CI/CD, manifests)       |
|   -> Coverage and semantic verification reports                   |
|   -> Updates to the platform-divergence catalogue as new divergences surface      |
+-------------------------------------------------------------------+
                            v
+-------------------------------------------------------------------+
| PHASE 5. Hybrid Deployment & Cutover                              |
```

```
  |   Staging deployment; source and target coexist in production;     |
  |   traffic is shifted progressively; the source system is           |
  |   decommissioned when all its responsibilities are covered.        |
  |   -> Coexistence architecture; cutover plan; sunset schedule       |
  +--------------------------------------------------------------------+
```

Every arrow in the diagram is a concrete artifact hand-off, not a narrative transition.
The pipeline is iterative: a Phase-4 discovery can send work back to Phase 3
(new migration-phase spec), Phase 2 (a deferred decision becomes needed),
or Phase 1 (a new dimension or pattern is discovered).
A complete artifact inventory is given in Appendix B.

Phase 1 — Knowledge Extraction

Goal. Produce a complete, evidence-based model of the source system
and bind it to the business intent of the migration.
Everything downstream is derived from what Phase 1 finds.

Inputs. Source repository (treated as read-only), domain documentation,
subject-matter-expert interviews, recorded walkthroughs, production configuration samples,
operational logs, and any external specifications or contracts the system implements.

Outputs (three artifact families).

1. **Source architecture specifications** — one per dimension discovered in the source
   (see the dimension sub-sections below).
2. **Requirements document** — Mission / Goals / Premises / Constraints
   plus a Requirements Traceability Matrix (RTM).
3. **Platform-divergence catalogue** — behavioural divergences between the source and target platforms
   (40+ categories in the reference project; see
   Appendix F).
   The catalogue starts in Phase 1 and grows through every later phase.

Each output is self-contained and cross-references the source by relative path
and line number rather than duplicating source content.

1.1 Dimensions — what they are and what they are for

A *dimension* is any axis along which source elements are related in a way that
materially constrains migration. Every software system exposes its own set;
the framework does not fix the list in advance.
Dimensions serve four operational purposes across the pipeline:

Purpose	Used in phase	Effect
Knowledge extraction	Phase 1	Each dimension becomes a separate specification; together they form the complete source model
Phase ordering	Phase 3 (Appendix C)	The «driving dimension» chosen as primary determines the flow variant and phase sequence
Coverage verification	Phase 4	Coverage is reported *per dimension* — edges, entities, endpoints, hook sites, pipeline steps
Semantic verification	Phase 4 (Appendix G)	Integration-pipeline contracts are expressed as traversals of specific dimensions

Common dimensions

Real projects typically infer six to twelve dimensions; the catalogue in Appendix A lists twenty-plus, and Appendix B shows the concrete set produced for one worked example.

Dimension (examples)	Primary purpose it serves
Dependency graph between code elements	Phase ordering + coverage per call-site
Data-model / entity graph	Phase ordering (models before business logic) + schema coverage
Module / package graph	Phase ordering + build-order verification
File relationship graph (include / dynamic load)	File-level partitioning + coverage
Frontend / backend coupling graph	Semantic verification of cross-boundary contracts
Service dependency graph	Phase ordering for microservice decomposition
Event / message flow graph	Phase ordering + topic-by-topic cutover
Plugin and extension-point graph	Hook-site coverage + semantic parity for third-party integrations
Data-pipeline topology (sources → sinks)	Phase ordering for DAG migrations + pipeline-contract verification
Protocol state machine	Semantic parity for state transitions and timer behaviour
Configuration / feature-flag surface	Coverage of runtime toggles
Security surface (auth, crypto, session)	Semantic verification of trust boundaries
Concurrency and scheduling model	Semantic parity for ordering and deadline behaviour
Regulatory / compliance boundary	Coverage of mandatory controls
Hardware-abstraction layer	Phase ordering for embedded / IoT migrations
API contract surface (inbound / outbound)	Coverage and semantic parity for external consumers

On the source-file inventory. Because the file relationship graph is itself a dimension, there is no separate «table of files» artifact. The file-level view is one dimension specification among others, produced by the same inventory and analysis step.

1.2 Requirements document

The requirements document has four mandatory components. They are **abstract and generic**: the concrete name of the containing artifact depends on the governance model adopted. Most current agent tooling (including the widely-adopted spec-kit) defaults to Agile-flavoured containers; the framework is neutral over the choice — see Appendix B.1 for a short list of equivalents.

Component	Definition
Mission	Single-sentence terminal value answering *why* the migration exists
Goals	Concrete objectives that, if changed, change the solution type
Premises	Assumptions that must hold for the goals to be achievable
Constraints	Hard (violation = rejection) and soft (violation = penalty) boundaries

The four components are produced by combining bottom-up knowledge expansion (which surfaces implicit Premises and Constraints) with top-down intent inference (which identifies the Mission and freezes the Goals). The concrete prompt-engineering method is documented in the `requirements-extractor` skill; the framework cares only about the four-component output, not the procedure that produced it.

Phase 1 also produces a **requirements traceability matrix** linking every requirement to the dimension specification where it was discovered, the decision record(s) where trade-offs about it will be resolved (Phase 2), and the migration-phase specification(s) where it will be satisfied (Phase 3). The RTM is the primary device that keeps every later phase mutually consistent.

The requirements framework is itself replaceable

The four-component shape (Mission / Goals / Premises / Constraints) is the framework default because it fits nearly every established governance framework. Projects with strong prior art in a different tradition may substitute any equivalent quadruple. In practice even less-structured inputs work — a free-form narrative inferred by the LLM directly from source, docs, and SME input can stand in for the four-component document, as long as Phase 2 can still cite stable «why?», «what?», «assumes?» and «must hold?» elements.

Tradition	Terminal value	Frozen objectives	Assumptions	Constraints
Default (this document)	Mission	Goals	Premises	Constraints
RUP Vision Document	Vision	Objectives	Assumptions	Constraints
PMBOK Project Charter	Purpose	Objectives	Assumptions	Constraints
Lean canvases	Problem / Opportunity	Solution	Key metrics	Unfair advantages
INCOSE	Stakeholder need	System requirements	Interface assumptions	Regulatory constraints

The framework requires only that the substitute preserves three invariants:

- a **terminal value** that justifies all downstream work;
- a **set of frozen objectives** whose change forces re-planning;
- **auditable assumptions** and **auditable constraints** that feed Phase 2 decisions.

When the three invariants hold, the substitute slots in without further adaptation.
When they do not, the framework is either extended or the substitution is rejected as a Phase-2 decision.

Governance-model neutrality

The choice of governance model is itself a first-phase decision.
Different models use different artifact containers for the four components:

Governance model	Typical container for the four components
Agile	Project charter; sprint zero charter
Waterfall	Project definition plan plus work-breakdown structure
PRINCE2	Project brief plus project initiation document
PMBOK	Project charter
SAFe	Portfolio vision plus solution intent
RUP	Vision document
Regulated programme	Programme vision plus statement of architecture work
Bespoke	Project-specific container

The framework is neutral over the container; the four components are mandatory;
the container name and the surrounding governance artifacts are project-specific.

1.3 Platform-divergence catalogue

Deep research on the source and target platforms invariably surfaces patterns where the two behave differently despite superficially similar code. These patterns are the most common source of silent semantic defects during migration, and a project-specific catalogue of them pays for itself very quickly. See Appendix F for a representative taxonomy, and Appendix B for how a concrete catalogue was stored in one reference project.

Each catalogue entry records, at minimum:

- a stable code and name;
- the source-platform behaviour;
- the target-platform behaviour and where they diverge;
- a representative source / target code pair;
- an estimated frequency in the codebase;
- the recommended remediation.

The catalogue is a living artifact. It is seeded in Phase 1 from research and SME input, grows in Phase 4 as new divergences surface during semantic verification, and feeds every audit checklist thereafter.

Phase 2 — Decisions

Goal. Propose every non-trivial target-architecture and migration-strategy choice, accept each decision with the right stakeholders, and record the rationale so that the decision is auditable and re-openable if circumstances change.

2.1 Decision records

Each decision is captured as a formal record (MADR, ISO/IEC/IEEE 42010, Y-statements, or bespoke templates all work) with the shape: context, considered options, trade-off analysis, decision, consequences, confirmation criteria. An index (and where useful, a dependency graph between decisions) is kept alongside the records so that readers can navigate the full decision log.

2.2 Stakeholder acceptance

Decisions are not accepted by the framework — they are accepted by the people responsible for the outcome. Each decision record nominates the deciders and the parties consulted and informed. Typical roles are:

Role	Accountable for
Product team / product owner	Business impact; scope boundaries; feature keep / deprecate / enhance
Project architect	Target architecture coherence; flow variant; cross-decision consistency
Subject-matter experts (SME)	Domain-specific constraints; behavioural parity; regulatory boundaries
Security / compliance lead	Trust boundaries; credential handling; audit obligations
Operations / platform lead	Deployment topology; CI/CD; infrastructure-as-code; observability
Development lead	Implementability within team capability and timeline

Multi-party review is what gives Phase 2 its value; a decision accepted solely by the implementer is a design note, not a governance artifact.

2.3 Priority classes and acceptance modes

Priority	Scope	Accepted before
P0	Blocks all work (flow variant, target stack, data engine)	Phase 3 can begin
P1	Blocks a specific migration phase (ORM, auth, workers)	That migration phase
P2	Deferrable, no blocking dependency	The final release

Decisions may be accepted in deferred mode. A decision does not have to be closed during inception. The framework supports three acceptance modes explicitly:

- **Up-front.** P0 decisions are normally accepted in Phase 2 itself.
- **Just-in-time.** A P1 decision can remain in *proposed* state until the migration phase that first needs it; the migration phase's entry gate then requires its acceptance.
- **Parallel with implementation.** A decision about an incremental concern (logging, observability, i18n) may evolve alongside early implementation as evidence accumulates; the decision is accepted once the design stabilises.

The framework explicitly supports decisions that are rejected or superseded later as evidence changes; a superseded decision carries a forward link to its replacement so that the full history remains auditable.

2.4 Option generation and consistency

Option generation. Each decision must contain at least two evaluated options with trade-off analysis. Options can come from human architects, vendor proposals, prior-art research, or iterative LLM-driven candidate generation. Appendix B.3 describes one structured elimination-and-replacement protocol; the framework is agnostic to how options are sourced.

Consistency constraint. When the status of a decision changes (proposed -> accepted, deprecated, superseded), every document that references that decision must be updated atomically. Partial status updates are a leading source of multi-session contradictions and must be treated as a build-breaking defect. A concrete consistency checklist is captured in Appendix B.2.

Phase 3 — Target Specifications

Goal. Produce a single unified set of specifications that covers both the target architecture *and* the migration strategy — what to build, in what order, under which decisions, with which exit gates.

3.1 Unified spec set

The framework does not separate «target architecture specs» from «migration plan specs» — they are the same set of documents. Each migration phase is described by three artifacts that together answer every question a developer or reviewer needs:

- **What should this part of the target look like?** — target structure, interfaces, data contracts
- **What must this part preserve from the source?** — behavioural parity, invariants, edge cases
- **How do we get there?** — order of batches, dependencies, scope boundaries
- **When is it done?** — acceptance criteria, exit-gate checklist, coverage thresholds

> *Example (from the reference project — see Appendix B.4): the migration-phase spec for authentication answered all four questions in a single document — target structure (server-side session store), preserved behaviour (session-cookie compatibility with existing clients), order of batches (models → verifier → endpoints → tests), and exit gate (legacy-password upgrade path verified; zero coverage gaps in the auth dimension).*

The three artifact types are:

- a **migration-phase specification** — user-visible behaviour, functional requirements, acceptance criteria, success criteria, explicit scope and anti-scope
- a **migration-phase plan** — technical context, the decision records the phase depends on, the migration batches in dependency order, risk assessment, entry and exit gates
- a **migration-phase task list** — actionable steps, parallel markers, and cross-references to the source elements each step covers

The primary driver dimension (chosen in Phase 2 as the flow-variant decision) determines the order in which migration phases are sequenced. A catalogue of flow variants and the factors that favour each is given in Appendix C.

3.2 Iterative generation

Phase 3 is **not** a single big-bang planning exercise. Migration-phase specifications are generated iteratively, starting with the phases that are fully unblocked by already-accepted decisions, and looping back to Phase 2 whenever a deferred decision becomes needed:

```
                                        + ----------------------------+
                                        |   accepted deferred decision |
                                        +-----^-----------------------++
                                              |                       |
                                              |                       v
  +-------------+   +---------------+   +-------------+   +------------------+
  |  Phase 1    |-->|  Phase 2      |-->|  Phase 3    |-->|   Phase 4        |
  |  knowledge  |   |  decisions    |   |  generate   |   |   execute spec   |
  |  +          |   |  (P0 now;     |   |  next spec  |   |   + coverage     |
```

```
  |  platform-  |   |  P1/P2 may    |   |  set for    |   |   + semantic     |
  |  divergence |   |  be deferred) |   |  first      |   |   + integration  |
  |  catalogue  |   |               |   |  unblocked  |   |   + e2e tests    |
  +-------------+   +-------^-------+   |  phase      |   +---------+--------+
                            |           +------+------+             |
                            |                  |                    |
                            |                  v                    |
                            |           +-------------+             |
                            |           | next phase  |             |
                            +-----------+ needs       |<------------+
                                        | decision    |
                                        | not yet     |
                                        | accepted?   |
                                        +-------------+
```

The loop terminates when the accepted-decisions set plus the already-generated migration-phase specs cover every requirement in the RTM.

Illustrative order (from the reference project, P0/P1 only):

1. Generate the spec set for the first unblocked phase (often a walking skeleton or foundational slice).
2. Execute that phase through Phase 4.
3. Return to Phase 3 to generate the next phase's specs — at this point previously deferred decisions may become needed; that triggers a short loop back into Phase 2 to accept them.
4. Continue until every migration phase is complete.

Every spec cites the decisions it depends on so that the dependency is explicit and auditable.

3.3 Exit gates

Every migration phase has an explicit exit gate listing machine-checkable conditions: tests passing, coverage thresholds, semantic verification clean, referenced decisions accepted. The gate is a checklist, not a narrative.

Artifact Flow

The framework produces a small set of artifact kinds, each with a clearly-defined producer phase and consumer phases. The diagram below shows the producer/consumer relationships; concrete container names, file layouts, and per-project examples live in [Appendix B — Artifacts & Lifecycle](#).

```
PHASE 1: Knowledge Extraction
    produces  ->  Source architecture specifications
                  (one per discovered dimension; plus source-element inventory)
               +  Requirements document
                  (Mission / Goals / Premises / Constraints)
               +  Requirements Traceability Matrix (RTM)
               +  Platform-divergence catalogue (grows through Phases 2-5)
               +  SME knowledge capture (recordings, transcripts, sheets)

PHASE 2: Decisions
    consumes  <-  Source architecture specifications
```

```
                <- Requirements document
                <- Platform-divergence catalogue
    produces  ->  Decision records (one per non-trivial choice)
                +  Decision index / dependency graph

PHASE 3: Target Specifications
    consumes  <-  Requirements document
                <- Decision records (P0 required; P1/P2 may be deferred)
                <- Source architecture specifications
    produces  ->  Migration-phase specifications (one set per migration phase)
                +  Migration-phase plans
                +  Migration-phase task lists
                [iterative: one migration-phase spec set at a time;
                 deferred decisions trigger a return to Phase 2]

PHASE 4: Migration Execution (one pass per migration phase)
    consumes  <-  Migration-phase spec / plan / tasks
                <- Requirements document (RTM lookups)
                <- Decision records
                <- Source architecture specifications (per-dimension coverage)
                <- Platform-divergence catalogue (semantic-verification checklists)
    inner loop (per batch):
                   produce target code with traceability
                -> coverage analysis (per dimension)
                -> GAP RESOLUTION (generate new OR enhance existing)
                -> SEMANTIC VERIFICATION (line-by-line on high-risk elements)
                -> integration + end-to-end tests
    produces  ->  Target artifacts
                   (code, schema, IaC, CI/CD, deployment manifests)
                +  Traceability links (source + specs + decisions)
                +  Coverage report + semantic-verification report
                +  New entries appended to the platform-divergence catalogue
                +  Session handoff note (-> next Phase-4 session)

PHASE 5: Hybrid Deployment & Cutover
    consumes  <-  Target artifacts (from Phase 4)
                <- Source system still in production
    produces  ->  Coexistence architecture
                +  Traffic-shift plan; cutover plan; sunset schedule
                +  Operational reports during coexistence

GOVERNING FILES (produced once; referenced from every phase):
        AGENTS.md              -> agent instructions (cross-tool OSS standard)
        Governing Principles   -> ordered project principles
        Skill library          -> reusable prompt-engineering methods
```

Where each kind of artifact lives

Artifact kind	Produced in	Consumed by	Concrete location (see Appendix B)
Source architecture specifications	Phase 1	Phases 2, 3, 4	Project-specific; typically a `specs/architecture/` tree
Requirements document	Phase 1	Phases 2, 3, 4	Project-specific container (charter, vision, PID, etc.)
Requirements Traceability Matrix	Phase 1	Phases 2, 3, 4	A table inside or beside the requirements document
Platform-divergence catalogue	Phase 1+	Phases 2, 3, 4	Project-specific; typically beside the source architecture specs
SME knowledge capture	Phase 1	Phases 2, 4, 5	Project-specific; often under `docs/` or a media library
Decision records	Phase 2	Phases 3, 4, 5	Project-specific; typically a `docs/decisions/` tree
Migration-phase specs / plans / tasks	Phase 3	Phase 4	Project-specific; typically a `specs/NNN-<phase>/` tree
Target artifacts + traceability	Phase 4	Phase 5	Target repositories — one or many, per the accepted topology
Coverage and semantic reports	Phase 4	Phase exit gates	Project-specific; emitted as CI/CD artifacts
Coexistence / cutover plan	Phase 5	Operations	Project-specific; alongside deployment manifests
AGENTS.md (+ governing principles)	Once	Every phase	Repository root (cross-tool OSS standard)

Governance file

The framework requires one file at the repository root that declares agent instructions and ordered project principles. The cross-tool open-source standard for this file is **AGENTS.md**. All governing principles referenced throughout this document — *behavioural parity*, *source traceability*, *incremental verification*, and so on — live inside AGENTS.md unless the project chooses a different container. Alternative containers (separate principles documents, framework-specific conventions) are listed in Appendix I.

Phase 4 — Migration Execution

Goal. Turn each migration-phase spec into verified target artifacts.
Phase 4 runs once per migration phase from Phase 3. Each run proceeds in a fresh working session so that context is clean and the session handoff artifact is the only carrier of cross-phase state. A run may emit artifacts into any number of target repositories — single-repository, one-to-many, and many-to-many migrations are all natural outputs of the same loop.

```
For each batch in the migration phase:
  (a) Produce target artifacts with traceability links
  (b) Structural coverage analysis  ->  gap report
  (c) Gap resolution                ->  generate new OR enhance existing
  (d) Semantic verification on high-risk elements
  (e) Build / static analysis / unit tests
  (f) Integration and end-to-end tests at current scope
  (g) If any step fails -> narrow scope, return to (a)
  (h) Otherwise -> next batch

Migration-phase exit:
  - Coverage report shows zero unmatched in-scope elements
  - All automated tests green (unit + integration + end-to-end)
  - All decision records the phase depended on are in "accepted" status
  - New platform-divergence catalogue entries merged
  - Session handoff note written for the next session
```

Coverage and semantic checks are executed *within* each batch of the loop, not at the end of the project. This is how defects are caught while context is hot rather than during late-stage integration. Integration and end-to-end tests are part of the exit gate, not a separate later stage.

Multi-session, multi-agent execution. The loop is designed for execution across many working sessions and, where beneficial, across multiple cooperating agents (reviewer + author, parallel sub-agents per batch, separate agents for coverage vs semantic verification). Session continuity and agent orchestration are agentic-system concerns — each AI-coding platform provides its own mechanism (handoff notes, persistent memory, delegation protocols). The framework only requires that whatever mechanism is used preserves the phase artifacts (specs, plans, tasks, decisions, traceability) between sessions.

The three core mechanisms that make this loop auditable — Traceability, Coverage Validation, and Gap Resolution — are described next.

The Core Mechanism: Traceability, Coverage, and Gap Resolution

Why AI models miss elements of the source

Even with large-context models capable of loading an entire repository, source elements are systematically missed during migration:

- **Attention dilution.** Long source files cause attention to concentrate on prominent elements while skipping less prominent ones.
- **False coverage.** File-level correspondence claims coverage of an entire file while actually covering only a subset of its elements.
- **Name collision.** Elements with common names are matched by name alone across unrelated files (a naive global-name validator makes this failure invisible).
- **Cross-session drift.** Work done in session N may conflict with assumptions from session $N - k$.

Structural coverage analysis reveals these gaps directly, and semantic verification then detects whether the covered code actually preserves behaviour.

Traceability

Every meaningful target element (function, class, method, model, route, constant, schema object, pipeline stage, configuration key) carries two kinds of link:

- **Source link** — one or more references to the source element(s) from which the target element was derived, together with a categorical link type (see below).
- **Specification / decision link** — one or more references to the phase specification, dimension specification, or decision record under which the target element was produced.

The second kind of link makes the target *doubly* traceable: it is possible to ask *which source shaped this code* and *which decision justified this shape*.

Source link types, in order of specificity:

Type	Meaning
Direct	Source element maps to a single target element
Method-level	Source method maps to target method within a class
File-level	Target module aggregates one source file
Multi-source	Target element combines logic from multiple source files
Schema-level	Target model derived from a schema definition
Inferred	Target code adapted from source patterns; no direct equivalent
New	Genuinely new code; no source equivalent
Eliminated	Source intentionally not ported (dead, deprecated, or superseded)

Traceability links can be recorded in several ways.
The representation is an implementation choice; the validator only requires that links be programmatically parseable. Representation options are described in Appendix D.

Structural Coverage Analysis and Gap Resolution

A common misconception is that a traceability-driven pipeline works like a transpiler — feeding source code line-by-line into an AI agent and receiving target code back.
It does not. The pipeline works for any migration style, including full architectural redesigns (monolith -> microservices, server-rendered -> SPA, SQL triggers -> stream processing, single repository -> many repositories).

The key insight:
traceability is used to detect what has not yet been accounted for;
it is never used to constrain how migration is done.
The AI agent is free to redesign, restructure, and re-architect the target system in whatever way the accepted decisions prescribe. Traceability only tracks which source elements have been accounted for.

```
                    SOURCE CODEBASE
                (all elements inventoried)
                           |
                    +------v-------+
                    |  COVERAGE    |
                    |  VALIDATOR   |
```

```
                    |              |
                    |  compares    |
                    |  source      |
                    |  inventory   |
                    |  against     |
                    |  target      |
                    |  traceability|
                    |  graph       |
                   +------+-------+
                          |
          +----------------+----------------+
          |                                 |
     ACCOUNTED FOR                    GAP REPORT
     (covered U eliminated)           (unmatched or low-confidence
          |                            source elements)
          |                                  |
          |                          +-----v-----+
          |                          | AI AGENT  |
          |                          | interprets|
          |                          | each gap  |
          |                          | and:      |
          |                          |  - adds a |
          |                          |    new    |
          |                          |    target |
          |                          |    element|
          |                          |  - or     |
          |                          |    enhances|
          |                          |    an     |
          |                          |    existing|
          |                          |    element|
          |                          |  - or     |
          |                          |    corrects|
          |                          |    a      |
          |                          |    previous|
          |                          |    one    |
          |                          +-----+-----+
          |                                  |
          |                          emit target edit +
          |                          traceability link
          |                                  |
          +----------------+----------------+
                          |
                   +------v-------+
                   | RE-VALIDATE  |
                   | coverage     |
                   | increases    |
                   | monotonically|
                   | until 100%   |
                   +--------------+
```

Worked example. Suppose the source is a monolith and the target is a set of microservices.
The validator reports that a source function has not been accounted for.
It does *not* instruct the agent to translate those lines.
Instead the agent:

1. reads the source function to understand its business role;
2. locates the correct target service per the accepted architecture;

3. inspects what the service responsibilities already cover;
4. implements the missing responsibility in the target idiom,
   or enhances an already-implemented function when coverage analysis shows that the existing implementation is incomplete, inconsistent with a new dimension, or contradicted by a later-accepted decision;
5. adds a traceability link so the validator treats the source element as accounted for.

This generate-*or*-enhance property is the decisive advantage of coverage-driven migration over direct translation and over purely agentic approaches.
In direct translation, once a target element is produced, further source analysis cannot correct it without re-running the whole translation.
In purely agentic approaches, there is no categorical signal telling the agent *which* existing element requires correction. Coverage analysis provides that signal: a source element whose current coverage is weak, inconsistent, or contradicted becomes an input to the next gap-resolution cycle regardless of whether target code already exists for it. The framework therefore supports gradual enhancement and correction of already-written code, not just addition of new code — which is what makes correct migration and modernisation possible at scale.

The mechanism applies equally well to:

- **Near-identical translations** — most elements map 1:1.
- **Framework migrations** — structure changes, elements mostly map.
- **Full redesigns** — source elements scatter across many targets; traceability tracks which are accounted for.
- **Technology shifts** — source units become fundamentally different target units (triggers -> streaming jobs, stored procedures -> services, monolith functions -> microservice endpoints).
- **Repository topology shifts** — one-to-many (decomposition), many-to-one (consolidation), many-to-many (reshaping).

The Coverage Validator

A source-agnostic validator performs four steps:

1. **Source inventory.** Parse every source artifact to capture an exact element list. The parsing technology is an implementation choice (agent-built regular expressions, language server indexes, or AST parsers); options and trade-offs are described in Appendix E.
2. **Target scanning.** Parse every target artifact for traceability links.
3. **Strict matching.** A source element is *covered* only when the target traceability link references the *same source location*. There is no global-name fallback; a target element linked to one source file does not cover a same-named element in a different source file.
4. **Dimension cross-check.** Verify that dependency edges, entity references, and integration points from the dimension specifications are present in the target.

Each source element falls into exactly one of three categories:

Category	Definition
Covered	Target code with a traceability link pointing to this source element
Eliminated	Documented as dead, deprecated, or superseded; not ported
Unmatched	No target equivalent — a gap that must be resolved before the phase exits

Unmatched elements feed Gap Resolution. The loop continues until
coverage reaches 100% of in-scope elements (eliminated + covered = total)
or until the phase exit gate is explicitly waived with a documented exception.

Semantic (Behavioral) Verification

Structural coverage proves that every source element has been accounted for.
It does not prove that the target behaves correctly.
A target element can be structurally covered but semantically wrong.

Semantic verification compares source / target element pairs according to a depth tier:

Tier	Criteria	Depth
1	Complex logic, many branches, cross-cutting callers	Line-by-line with source quoting
2	Moderate complexity	Block-level with spot checks
3	Simple wrappers, accessors	Signature and return type
4	Data models, schemas	Field-by-field

Tier-1 reviews quote raw source lines from both files.
Summarised reviews without source quotes systematically miss structural defects
(misindented loop bodies, dropped branches, inverted conditions).
A concrete, representative discrepancy taxonomy is given in Appendix F.

Use of the platform-divergence catalogue

Semantic verification reads from the platform-divergence catalogue
introduced in Phase 1.3:
the checklist for each element is the set of catalogue entries known to
apply to that element's source pattern. New divergences discovered during
verification are appended to the catalogue so that every subsequent migration
phase benefits from them.

Integration-pipeline verification

Per-element verification catches per-element defects.
It does not catch defects that emerge when data flows through multiple elements
with incompatible assumptions — for example, an identifier constructed one way in
function *f* but looked up differently in function *g*.

The framework defines a small number of end-to-end integration pipelines
and verifies each as an ordered contract: boundary-by-boundary,
the data shape produced at step k must match the shape consumed at step $k + 1$.
Appendix G describes the mechanism in detail.

Functionality Adjustment

Goal. Align migration scope with business needs *before and during* implementation, using the same artifact-driven pipeline.

The framework provides three integration points for product and subject-matter-expert input:

1. **Feature scope review** (during Phase 1). Product confirms which features are kept as-is, enhanced, deprecated, or eliminated. The outcome is a feature scope matrix referenced by all later phases.
2. **SME knowledge ingestion** (any time). Recordings, screenshots, spreadsheets, and walkthroughs of the working source application are ingested, transcribed, and used to generate end-to-end test scenarios and to discover features not visible from source code analysis alone.
3. **Sampled SME verification** (during each phase). Randomly sampled source / target element pairs are reviewed by SMEs. This catches divergences that no automated check can detect (domain-specific behaviour, undocumented business rules, cosmetic expectations).

Phase 5 — Hybrid Deployment & Cutover

Goal. Put enough of the target into a real deployment — staging first, then production alongside the source — to exercise it under realistic load and data, progressively shift responsibility, and decommission the source system when every responsibility is covered.

For most systems a big-bang cutover is prohibitively risky; source and target must coexist for some period. The framework treats the coexistence architecture itself as a decision, not a default. The pipeline is flexible enough to span the full range of project scales and repository topologies.

Release modes

Release mode	When appropriate	Consequence
Big-bang, single cutover	Small and mid-scale projects; full regression coverage feasible	Entire migration verified in one release; shortest timeline
Gradual coexistence	Large systems where full cutover risk is unacceptable	Source and target run side-by-side for months or years with a documented sunset
Incremental traffic shift	Systems with a gateway, service mesh, or feature-flag layer	A percentage of traffic is routed to the target; percentage increases over time
Read-first / write-later	Data-heavy systems where writes are the highest risk	Target serves reads while source remains authoritative for writes, then flipped
Parallel-run comparison	Pipelines, reports, or computations where output correctness is observable	Both systems process the same inputs; outputs are diffed until convergence

Output topology

The framework is **not** a repository-to-repository translator.
It is a knowledge-to-artifacts pipeline whose output is whatever set of repositories, CI/CD pipelines, infrastructure-as-code (IaC) modules, and deployment manifests the accepted architecture prescribes. The Phase-1 dimensions and the Phase-2 decisions together determine the target topology.

Source topology	Target topology	Typical migration
1 repository	1 repository	Language or framework migration; near-identical shape preserved
1 repository	*N* repositories	Monolith decomposition into services; one repo per service + shared libs
N repositories	1 repository	Consolidation of legacy component repositories into a single modular target
N repositories	*M* repositories	Full reshaping — service boundaries redrawn, new shared libraries extracted
1 repository	1 repo + IaC repo	Legacy app modernised with a separate operations / infrastructure repository

For every non-trivial output topology, the pipeline also produces:

- per-repository CI/CD configuration
- infrastructure-as-code modules (Terraform, CloudFormation, Pulumi, Kubernetes manifests, or equivalents)
- service-mesh / gateway configuration when applicable
- data-migration scripts with rollback plans (see Appendix H)
- observability configuration (logs, metrics, traces) consistent across repositories

Representative coexistence patterns

Pattern	Mechanism
Web applications	Shared database; API gateway; compatibility shim
Microservices	Service-by-service cutover; contracts preserved at each step
Protocol libraries	Linked side-by-side; conformance tests against both
Data pipelines	Parallel runs on a data subset; output equivalence check
Event-driven systems	Dual consumers during transition; producer-by-producer switchover
Embedded / firmware	Dual-image slots; staged rollouts with automatic rollback

Every shim, dual-path, or bridge is tracked as a removal item with a documented sunset.

Applicability Across System Types

The pipeline is technology-agnostic. What changes between applications is the set of dimensions discovered in Phase 1, the decisions taken in Phase 2, and the flow variant chosen in Phase 3.
Appendix A catalogues dimensions observed across system types;

Appendix C catalogues flow variants.

The three core mechanisms (Traceability, Coverage Validation, Gap Resolution) apply unchanged to language migrations, framework shifts, monolith decompositions, CLI ports, desktop-to-web moves, network-protocol library ports, data pipeline rewrites, stored-procedure extractions, and embedded / IoT modernisations.

Benefits

The framework combines several established practices (decision records, requirements traceability, spec-driven development) with a small number of fundamental contributions that change what AI-assisted migration can do.

Fundamental contributions

- **Completeness is a measurable property, not a claim.**
  Traditional AI-assisted migration reports progress as «N% done» with no independent check. Here, every source element is in exactly one of three states (covered / eliminated / unmatched), and the phase exit gate rejects the migration phase until the count of unmatched in-scope elements is zero.
  «Done» becomes a programmatically verifiable predicate.
- **Dimensional coverage, not file coverage.**
  Coverage is reported *per dimension* — one report for the call graph, one for the entity graph, one for the event flow, and so on. A target that covers 95% of source functions but misses 40% of hook sites is flagged immediately, rather than shipping with a silent feature gap. This replaces single-number coverage metrics with a vector that reflects actual structural risk.
- **Generate-or-enhance, driven by coverage signals.**
  Unmatched or low-confidence coverage on an already-implemented target element triggers *correction or enhancement of that element*, not only generation of new code. This breaks the one-shot ceiling of transpiler-style migration (where an element once produced cannot be improved without re-running the whole translation) and the «no categorical signal» failure mode of purely agentic approaches (where nothing tells the agent *which* element to revisit).
  The same cycle gracefully absorbs late-breaking source discoveries and newly-accepted decisions.
- **Unified target-architecture and migration-strategy specification.**
  Phase 3 produces one spec set, not two. The question «what should this look like?» and the question «how do we migrate to it?» are answered in the same document, which eliminates the drift that accumulates when architecture and migration plans are maintained separately.
- **Platform-divergence catalogue as a living artifact.**
  A project-specific catalogue of source/target behavioural divergences is seeded in Phase 1 and grown in every later phase. It transforms semantic verification from ad-hoc review into a checklist-driven audit with falsifiable entries, and prevents the same class of error from recurring across migration phases.
- **Iterative, deferrable decisions bound to migration phases.**
  P1/P2 decisions can be accepted just-in-time, or in parallel with implementation, without blocking progress on unrelated phases. The RTM and the decision index keep the dependency graph explicit; a Phase-4 discovery can legitimately re-open a Phase-2 decision without invalidating already-completed migration phases.

Standard benefits (delivered as side effects of the contributions above)

- **Explicit trade-offs.** Every non-trivial choice is recorded as a decision with analysis, reviewable by non-specialist stakeholders and re-openable when business needs change.
- **Early error detection.** Errors are caught within the same session they are introduced, not during final integration.
- **Test-coverage uplift for legacy systems.** The target ends with richer automated test coverage than the source ever had — migration becomes a quality event, not a risk event.
- **Flexible output topology.** The pipeline produces whatever set of repositories, CI/CD pipelines, infrastructure-as-code, and deployment manifests the accepted architecture prescribes — single-repo, one-to-many, many-to-one, or many-to-many.

Appendices

All appendices are either catalogues, illustrative examples, or implementation notes.
Each real project will instantiate its own versions of these tables.
Nothing in the appendices is prescriptive.

Appendix A — Dimension Catalogue

Dimensions commonly discovered during Phase 1.
Every real project should infer its own set.

Structural dimensions (common to most systems)

Dimension	Captures	Migration impact
Call graph	Which elements call which others	Dependency order for migration
Data model / entity graph	Tables, schemas, relationships, clusters	Models required before business logic
Module / package graph	Import and include chains between packages	Build order; circular-dependency risk
File relationship graph	Static include and dynamic load edges between files	File-level partitioning
Service dependency graph	Inter-service calls	Distributed-system migration order
Event / message flow graph	Producers, topics, subscribers	Event-driven migration order
Infrastructure dependency	Services <-> infrastructure (DB, cache, queue)	Cloud / platform migration

Domain-specific dimensions (examples)

Dimension	Example system types
Frontend / backend coupling	Web apps with server-rendered HTML or SPAs
Plugin / extension-point graph	Applications with hook points and extension APIs
Protocol state machine	Network-protocol implementations
Data pipeline topology	ETL, stream processors, trigger chains
Configuration / feature-flag surface	Systems with extensive config or feature flags
Security surface	Auth flows, encryption, session management
Concurrency and scheduling model	Multi-threaded, async, event-driven
API contract surface (inbound)	Public APIs with external consumers
API contract surface (outbound)	Integrations with third-party services
Regulatory / compliance boundary	Regulated industries (finance, health, defence)
Hardware abstraction layer	Embedded / IoT
Data warehouse / OLAP cubes	Analytics platforms
File-based persistent state	Legacy systems with disk-backed state
Search index topology	Systems with dedicated search infrastructure
Object-storage key namespace	Systems with substantial blob storage
Scheduler / cron graph	Systems with time-driven background work
Multi-tenant isolation boundary	SaaS platforms
Observability topology	Systems with custom metrics / log pipelines

Applicability matrix

System type	Typical primary dimension	Typical secondary dimensions
Web application (MVC)	Entity graph	Call graph, frontend coupling
Server-rendered app	Frontend / backend coupling	Entity graph, template system
Desktop application	UI event model	Call graph, persistence
CLI tool	Call graph	Argument parsing, I/O model
Data processing pipeline	Pipeline topology	Data model, checkpoint semantics
Network protocol library	Protocol state machine	Timer semantics, packet encoding
Microservices	Service dependency graph	API contracts, message topics
Stored-procedure system	Pipeline topology	Entity graph, transaction boundaries
Event-driven architecture	Message topics	Service graph, event schema
Embedded / IoT	Hardware abstraction	Call graph, memory model
Analytics warehouse	OLAP cube / DAG topology	Data model, scheduler graph
SaaS multi-tenant platform	Tenant isolation boundary	Service graph, data-model graph

---

Appendix B — Artifacts & Lifecycle

This appendix gathers everything about the artifacts the framework produces:
the per-phase lifecycle they follow, a concrete inventory from one reference project,
and the decision-evaluation protocol that feeds Phase 2.

B.1 Spec-driven lifecycle

Every phase flows through the same six-step lifecycle:

```
Principles -> Specification -> Planning -> Tasks -> Execution -> Validation
```

Step	Artifact	Content
Principles	Governing principles	Ordered project-level rules (behavioural parity, traceability, etc.)
Specification	Phase specification	Behaviour, functional requirements, acceptance criteria, success criteria
Planning	Phase plan	Technical context, batches, risks, entry and exit gates, decision dependencies
Tasks	Phase task list	Checkboxed actions, parallel markers, source cross-references
Execution	Target artifacts	Code, schema, pipelines, infrastructure-as-code, CI / CD
Validation	Exit gate	Tests green, coverage validated, semantic verification clean

B.2 Consistency rule

When any status, decision, or phase changes, all referencing locations must be updated atomically: decision record, decision index, requirements document, dimension specifications, session memory. Partial updates create contradictions that compound across sessions. The consistency check is a machine-checkable checklist that enumerates every document that references the affected status field.

B.3 Iterative decision evaluation

For critical decisions (typically P0 and some P1), the framework can generate multiple candidate options and refine them through structured comparison:

1. Generate three initial candidates, each optimising for a different trade-off axis (time-to-market, technical excellence, behavioural fidelity).
2. Review each candidate independently for fatal flaws and unaddressed constraints.
3. In each round, eliminate the weakest candidate (fewest favourable pairwise comparisons) and replace it with a new candidate addressing flaws identified in prior rounds.
4. Repeat with two survivors plus one new challenger. Each round has constant cost: exactly three candidates, three pairwise comparisons.
5. Iterate until candidates converge or diminishing returns are observed.

This is one way to generate options; they may also come from human architects, vendor proposals, or prior-art research. The framework requires at least two evaluated options per decision but is agnostic to how they are produced. Pattern files and runtime harnesses for this protocol live in `.agents/skills/adversarial-thinking/` and `.agents/skills/adversarial-self-refine/`.

B.4 Reference project — artifact inventory

One concrete end-to-end application of the framework produced the artifact set below. The inventory is indicative, not prescriptive; each real project produces a different set depending on its discovered dimensions and accepted decisions.

Summary metrics.

Metric	Value
Source lines of code	~18,600
Source files	138
Database tables	35 (31 active, 4 deprecated / eliminated)
Plugin hooks	24
Inbound API endpoints	17 REST, 40+ RPC
Working sessions	19
Dimension specifications	15 (architecture root + per-dimension documents)
Phase specifications	6
Architecture decisions	19
Automated tests at completion	1,474 (unit + integration + end-to-end)
Platform-divergence entries catalogued	40 categories, 600+ affected call sites
Integration pipelines verified	8
Final structural coverage	100% (458 / 458 in-scope elements)
Eliminated source elements	27+
Security modernisations during migration	Password-hash upgrade, credential encryption, prepared statements, CSRF, security headers

Phase 1 — Source architecture specifications.

Path	Content
specs/architecture/01-architecture.md	Application layering, patterns, request lifecycle
specs/architecture/02-database.md	Entity / schema dimension
specs/architecture/03-api-routing.md	Inbound API contract surface
specs/architecture/04-frontend.md	Frontend / backend coupling
specs/architecture/05-plugin-system.md	Plugin / extension-point graph
specs/architecture/06-security.md	Security surface
specs/architecture/07-caching-performance.md	Caching, concurrency, scheduling
specs/architecture/08-deployment.md	Deployment topology
specs/architecture/09-source-index.md	File relationship graph (roles, dependencies, annotations)
specs/architecture/10-migration-dimensions.md	Call-graph communities; cross-dimension synthesis
specs/architecture/11-business-rules.md	Business-rule inventory
specs/architecture/12-testing-strategy.md	Test-category matrix
specs/architecture/13-decomposition-map.md	Source-element -> target-module decomposition
specs/architecture/14-semantic-discrepancies.md	40-category discrepancy taxonomy, semantic traps, pipeline contracts
specs/architecture/15-sme-review.md	Feature inventory from SME walkthrough

Phase 1 — Requirements document.

Path	Content
specs/architecture/00-project-charter.md	Mission, Goals, Premises, Constraints, requirements traceability matrix (Agile container)

Phase 2 — Decision records.
`docs/decisions/0001-migration-flow-variant.md` through `0019-preferences-modal-pattern.md`, plus `docs/decisions/README.md` (index and dependency graph) and `docs/decisions/compliance-review-response.md` (cross-decision review).

Phase 3 — Migration-phase specifications.

Path	Scope
specs/001-foundation/{spec,plan,tasks}.md	Models, auth, DB bootstrap, app factory
specs/002-core-logic/{spec,plan,tasks}.md	Feed parsing, counter cache, filters, labels, sanitisation
specs/003-business-logic/{spec,plan,tasks}.md	Preferences CRUD, digests, OPML
specs/004-api-handlers/{spec,plan,tasks}.md	17 API operations, authentication guards, tree-building algorithms
specs/005-semantic-verification/{spec,plan,tasks}.md	40-category taxonomy, 105+ remediations, 598 tests, 0 unresolved gaps
specs/006-deployment/{spec,plan,tasks}.md	CI, coverage gate, Docker / Compose / reverse proxy, data migration

Governing files (produced once, referenced from every phase).

Path	Role
AGENTS.md	Cross-tool agent instructions (OSS standard)
Ordered principles (in AGENTS.md or standalone)	Project-level rules consulted at every phase gate
`.agents/skills/`	Reusable prompt-engineering methods (requirements extraction, etc.)

B.5 Representative artifact shapes

Decision record excerpt (one of several equivalent templates):

```
# 0002 — Select Target Web Framework

## Considered Options

1. Framework A — closest match to source handler dispatch; native session and CSRF
2. Framework B — modern async; no built-in sessions; requires client-side changes for CSRF
3. Framework C — full-featured but over-engineered for this codebase

## Decision: Framework A

Rationale: minimises translation distance; preserves client-server contract.
```

Traceability link examples (representation: inline comment):

```
# Direct match
# Source: <source path>:<qualified name> (lines 706-771)
# Specs:  specs/architecture/06-security.md, specs/001-foundation/spec.md
# Decisions: 0008-password-migration.md

# Multi-source
# Source: <source1>:<name> (lines 45-120)
#       + <source2>:<name> (lines 200-450)
# Specs: specs/architecture/02-database.md, specs/architecture/03-api-routing.md
```

```

# Inferred
# Inferred from: <source pattern>; no direct source equivalent
# Specs: specs/001-foundation/spec.md
# Decisions: 0007-session-management.md

# New (no source equivalent)
# New; no source equivalent — schema migration infrastructure
# Specs: specs/006-deployment/spec.md

# Eliminated
# Eliminated: <source path>:<name> — removed per ADR 0003 (target uses single DB engine)
```

Appendix C — Flow Variants

The overall order of phases is itself an architecture decision.
Each project discovers its own optimal flow based on its dimensions.
The list below is illustrative — real projects frequently combine or adapt entries.

Variant	Strategy	Typical scenario	Primary risk
Entity-first	Models / schemas -> logic -> handlers	Database-heavy, stable data model	Long time to first runnable code
Call-graph-first	Entry points -> fill in dependencies	API-heavy systems, microservices	Stub accumulation
Vertical slice	End-to-end features, one at a time	Large team, parallel workstreams	Cross-cutting concerns extracted too late
Minimal runnable first	Smallest working system, then expand	Small team, layered architecture	More upfront planning
One-dimension-per-pass	Each pass addresses one dimension	Large team with specialists	High coordination overhead
Protocol-structure-driven	Packet structures, state machines first	Network protocol libraries	—
Service-graph-driven	Leaf services first, work inward	Microservice decomposition	—
DAG-topology-driven	Sinks first, then transforms, then sources	Data pipeline migrations	—
Contract-first	API contracts first, implementation second	Systems with external consumers	—
Message-topic-driven	By event type or topic cluster	Event-driven architectures	—
Risk-first	Highest-risk components first	Safety-critical modernisations	—
Compliance-first	Regulated boundaries first	Finance, health, defence	—
Repository-split	One dimension per target repository	Monolith-to-microservices	Cross-repository CI complexity

Factors that favour each variant

Factor	Favours
Small team	Minimal runnable first (fast feedback)
Large team	Vertical slices or one-dimension-per-pass
External API consumers	Contract-first
Time-to-market pressure	Minimal runnable first
Regulatory constraints	Entity-first or compliance-first (stability)
High safety stakes	Risk-first
Repository topology shift	Repository-split or service-graph-driven

Appendix D — Traceability Link Representations

The representation of traceability links is an implementation choice,
not part of the framework normative contract.
The validator only requires programmatic parseability. Options include:

Representation	Description	Trade-offs
Inline source comments	Structured comment blocks above each target element	No extra files; travels with the code; clutters code
Sidecar index files	JSON / YAML / TOML index mapping target paths to source + decision metadata	Target code stays clean; requires index discipline
Dedicated graph database	Nodes for source and target elements; edges for traceability links	Rich querying; infrastructure overhead
Commit-level metadata	Traceability encoded in commit messages or Git notes	History-preserving; requires VCS tooling
Specification-side mapping tables	Mapping tables stored inside phase specifications	Human-readable; coupling to specs
Hybrid	Lightweight inline pointer + authoritative sidecar index	Best of both; duplication risk

The reference project used inline comments for source links and inline references
for specification and decision links. Projects with substantial existing tooling
(Bazel, Buck2, language-server ecosystems) may prefer sidecar indexes
or graph databases.

Appendix E — Source Inventory and Validator Implementation

The coverage validator needs an exact inventory of source elements with boundary information.
Several implementation strategies exist, in increasing order of rigour:

Strategy	Description	Best for
Agent-built regex sweep	The AI agent builds language-specific regexes and enumerates elements	Small codebases; prototypes
Language server / indexer	Use IDE-grade indexers to enumerate symbols	Languages with mature tooling
AST parsing	Parse each source file to a full syntax tree and extract element boundaries	Large, heterogeneous codebases
Compiler-produced symbol graphs	Extract symbols from compiler intermediate representations	Compiled languages with stable IR
Query over existing code-intelligence graph	Query an existing source-graph store (e.g. SCIP, stack-graphs, CodeQL)	Projects already instrumented

Naive strategies (name-based grep without file scoping) produce false-positive coverage; strict strategies (full AST with exact start and end boundaries and file-scoped matching) produce reliable coverage. The trade-off is engineering effort versus precision. The validator output shape is the same regardless of strategy: covered / eliminated / unmatched, with a citation for every claim.

Appendix F — Semantic Discrepancy Taxonomy

A real project will discover its own taxonomy. The pattern is that discrepancies cluster into a small number of categories (fewer than 50 in most systems), so a per-project catalogue pays for itself quickly. A representative 40-category taxonomy observed in the reference project (Appendix B) spans six domains:

Domain	Example categories
SQL semantics	Missing JOIN, wrong WHERE, missing ORDER BY, wrong LIMIT / OFFSET, missing owner-scope guard, SQL dialect remnant, transaction-boundary change, parameter type divergence
Type system	Language-specific truthiness, coercion rules, numeric-boundary off-by-one, null vs empty, array vs map semantics
Data flow	Identifier construction mismatch, content priority inversion, field truncation, timestamp validation, encoding normalisation
Session / state	Session state elimination, session fallback, config constant mapping, profile-aware queries, global-variable elimination, in-memory cache elimination
Return values	Wrong shape, wrong field names, return materialisation, error envelope, HTTP header omission, JSON response structure
Features / behaviour	Feature absent, hook argument mismatch, hook call-site missing, side-effect order, error-recovery model, DOM / parsing model, transactional semantics

Each taxonomy entry is a pattern with a code, a name, a description, a representative source / target pair, and an estimated frequency in the codebase. The taxonomy drives two things:

- semantic verification checklists applied to each element according to its tier;
- a remediation catalogue telling the agent how to rewrite each pattern correctly.

Appendix G — Integration Pipeline Contracts

Per-element verification catches per-element defects. Defects also emerge at element boundaries: data produced in one element does not match the shape consumed in the next.

The framework defines a small set of end-to-end integration pipelines and verifies each as an ordered contract. Each pipeline is a chain of steps; the contract for step k to step $k + 1$ is the exact shape of the data at that boundary, the invariants that must hold, and the side effects that must have completed.

Representative pipelines (from the reference project — see Appendix B):

Pipeline	Steps
Feed update	Schedule -> fetch -> parse -> sanitise -> dedup -> persist -> update counters -> fire hooks
Article search	Query build -> SQL -> pagination -> hydrate -> permission check -> response
API request	Authenticate -> dispatch -> validate -> handle -> shape response -> sequence echo
Authentication	Credential -> hash verify -> session create -> cookie -> hydrate state
Counter cache	Event -> invalidate -> recompute -> publish
OPML roundtrip	Import parse -> validate -> persist <-> export query -> serialise
Digest generation	Query due -> render -> send -> mark sent
Plugin lifecycle	Discover -> load -> initialise -> register hooks -> invoke hooks

Each pipeline contract is stored as a table of boundary shapes.
Verification walks the pipeline end-to-end, checking that neighbouring steps agree.

Appendix H — Data Migration and Transformations

Data migration covers all forms of persistent and streaming state —
not just relational databases. The approach depends on the data type.

Database migration

1. **Schema mapping.** Source schema analysed dimension-by-dimension;
   target models generated and reviewed.
2. **Transformation rules.** When schemas differ, transformations are documented
   as decisions with rollback plans.
3. **Subset migration** for development and testing. Seed data per entity cluster,
   foreign-key-ordered insertion, personally-identifiable-information anonymisation.
4. **Verification.** Post-migration row counts, foreign-key integrity, spot-check queries.

Other data types

Data type	Considerations
Message schemas	Schema versioning, consumer / producer compatibility, dead-letter handling
ETL pipelines / stored procs	Topology preservation, checkpoint / restart, idempotency, backfill
Data warehouse / OLAP cubes	Materialised-view recreation, aggregation equivalence, historical backfill
File-based state	Format conversion, directory structure, permission model
Search indexes	Index schema, re-indexing strategy, relevance-tuning validation
Object storage	Key namespace, metadata preservation, access-policy translation

Each data-migration type is documented as a decision with transformation rules,
a rollback plan, and verification criteria.

Appendix I — Agent Integration

This appendix gathers the two artifacts that connect the framework to the underlying AI agent: the governance file at the repository root, and the library of reusable skills.

I.1 Governance file — AGENTS.md and alternatives

AGENTS.md is the cross-tool open-source standard for AI-agent instructions, stewarded by the Linux Foundation's Agentic AI Foundation and honoured by most modern AI coding agents. A single AGENTS.md at the repository root is the recommended container for both agent instructions and the project's ordered governing principles.

Projects with a pre-existing governance convention may substitute an equivalent container, provided agents can find it. Common alternatives:

Container	Origin / convention
`AGENTS.md` (recommended)	Cross-tool OSS standard
`AGENTS.md` + separate `PRINCIPLES.md`	Larger projects that separate concerns
`.github/copilot-instructions.md`	GitHub Copilot convention
`CLAUDE.md`	Anthropic Claude Code convention
`.cursorrules` / `.cursor/rules/`	Cursor convention
`constitution.md` + `AGENTS.md` pointer	Spec-kit convention (Agile-flavoured)
Bespoke governance document	Regulated-industry programmes

The framework is neutral over the physical container — it only requires that principles be version-controlled, ordered, and cited by every phase that depends on them.

I.2 Skill library

The framework is implemented on top of a library of reusable skills.
Copies are maintained in `.agents/skills/` within this project for self-containment.

Skill	Purpose
`requirements-extractor`	Two-phase (bottom-up Chain-of-Knowledge -> top-down intent) Mission / Goals / Premises / Constraints derivation
`deep-research-t1`	Tier-1 deep research pipeline for grounding decisions in authoritative sources
`knowledge-management`	Chain-of-Knowledge saturation across a large corpus
`document-ingestion`	Ingestion of transcripts, video frames, spreadsheets into structured knowledge
`adversarial-thinking`	Multi-candidate divergent design with pairwise comparison
`adversarial-self-refine`	Independent reviewer / author loop for iterative output refinement
`continuation-and-handoff`	Multi-session continuity and checkpoint protocols
`selecting-pe-methods`	Selection of prompt-engineering method per task archetype

Skills are consumed by invoking them through the agent skill loader;
each skill is self-describing and includes termination conditions and anti-patterns.

Appendix J — Presentation Deck

A slide-per-section distillation of this document, intended as speaker notes
or a handout. Each `### Slide` heading below is one slide.
Diagrams are given in Mermaid; text slides use short theses only.

Slide 1 — Title

AI-Assisted Software Migration & Modernization

An auditable, measurable approach for LLM-driven migrations and modernizations.

What it solves

- Missed source functionality that traditional LLM-driven migrations silently drop
- Behavioral divergences between source and target that only show up in production
- Untracked decisions and untraceable target code that block audit and hand-off
- Locked-in output shape (single-repo ports, monolith → microservices splits, consolidations, and full reshapes are all supported by the same pipeline)

What you get

- Complete, reviewable source knowledge — the starting point for every decision
- Explicitly accepted architecture and migration decisions with named stakeholders
- Target code that is traceable back to source, to specs, and to decisions
- Progressive deployment alongside the source system until full cutover

---

Slide 2 — The four pillars

Four concepts turn AI-assisted migration from a one-shot translation into an **auditable, measurable modernization** process. Each pillar answers one hazard that LLM-driven migrations routinely fail on.

Pillar	Hazard it neutralizes	What stakeholders gain
① Dimensions	Migration planned on the wrong axis; blind spots in source	Structured source model; migration order that respects reality
② Traceability	Target code of unknown provenance; no way to review fidelity	**SMEs can review fast** — every target element cites source + spec + decision
③ Coverage Validation	«N% done» self-assessments; silent omissions	«Done» is a **measurable predicate** — per-dimension, auditable
④ Modernization-Gap Resolution	Errors only surface in production; target cannot be enhanced	**Confidence** — gaps are detected and the target is iteratively corrected or enhanced, not only generated

> **Phases run concurrently and can re-enter one another.**
> The five phases that follow are not strictly sequential. Work in Phase 4 routinely triggers short returns to Phase 2 (accept a deferred decision), Phase 3 (generate the next migration-phase spec), or Phase 1 (a new dimension or divergence is discovered). Phase 5 overlaps Phase 4 once enough target is ready to deploy alongside the source.

The next three slides zoom into each pillar:
Slide 3 — **Dimensions** · Slide 4 — **Traceability, Coverage, Gap Resolution**.

Slide 3 — Pillar ① Dimensions

A *dimension* is a structural axis along which **source architecture elements** are related. Dimensions are **discovered per project**, not prescribed, and they determine migration order, partitioning, and what «complete» means.

Examples (named by the relationship they capture):

- **Code** — caller ↔ callee relationships between functions, classes, modules
- **Data** — foreign-key and joinable-column relationships between tables
- **API** — client ↔ endpoint relationships (inbound and outbound)
- **Events** — producer → topic → consumer relationships
- **Plugins / hooks** — hook-site ↔ handler relationships
- **Protocol state** — transitions and timers between states
- **Security** — principal ↔ resource ↔ permission relationships
- **Configuration** — feature-flag ↔ code-path relationships

From the reference project (PHP → Python web application): Phase 1 discovered fifteen dimensions, each stored as a separate source architecture specification:

- `specs/architecture/02-database.md` — the data dimension (foreign-key relationships)

- `specs/architecture/05-plugin-system.md` — the plugin/hook dimension
- `specs/architecture/06-security.md` — the security dimension
- ...and twelve more, listed together in [Appendix B.4]

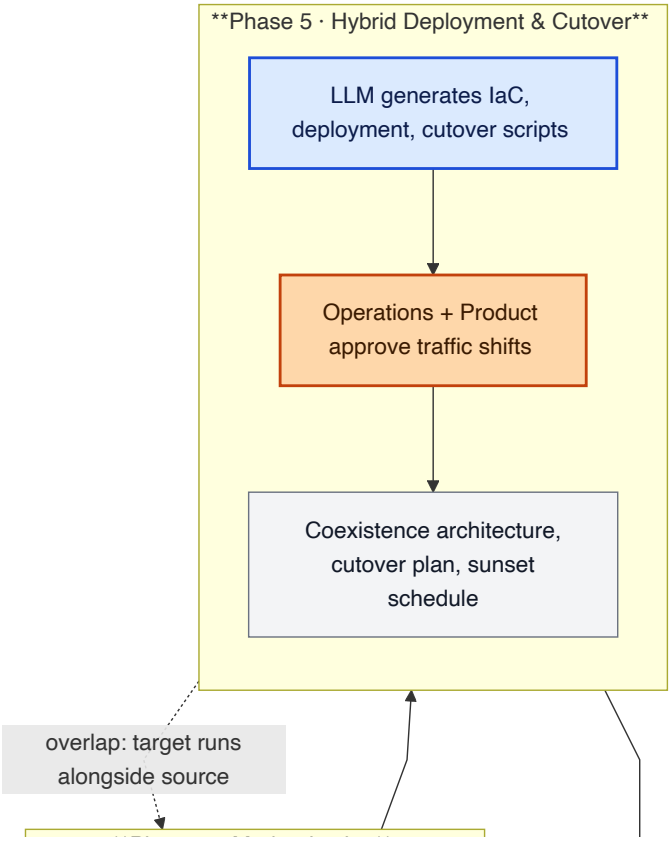
Slide 4 — Pillars ②③④ Traceability, Coverage, Gap Resolution

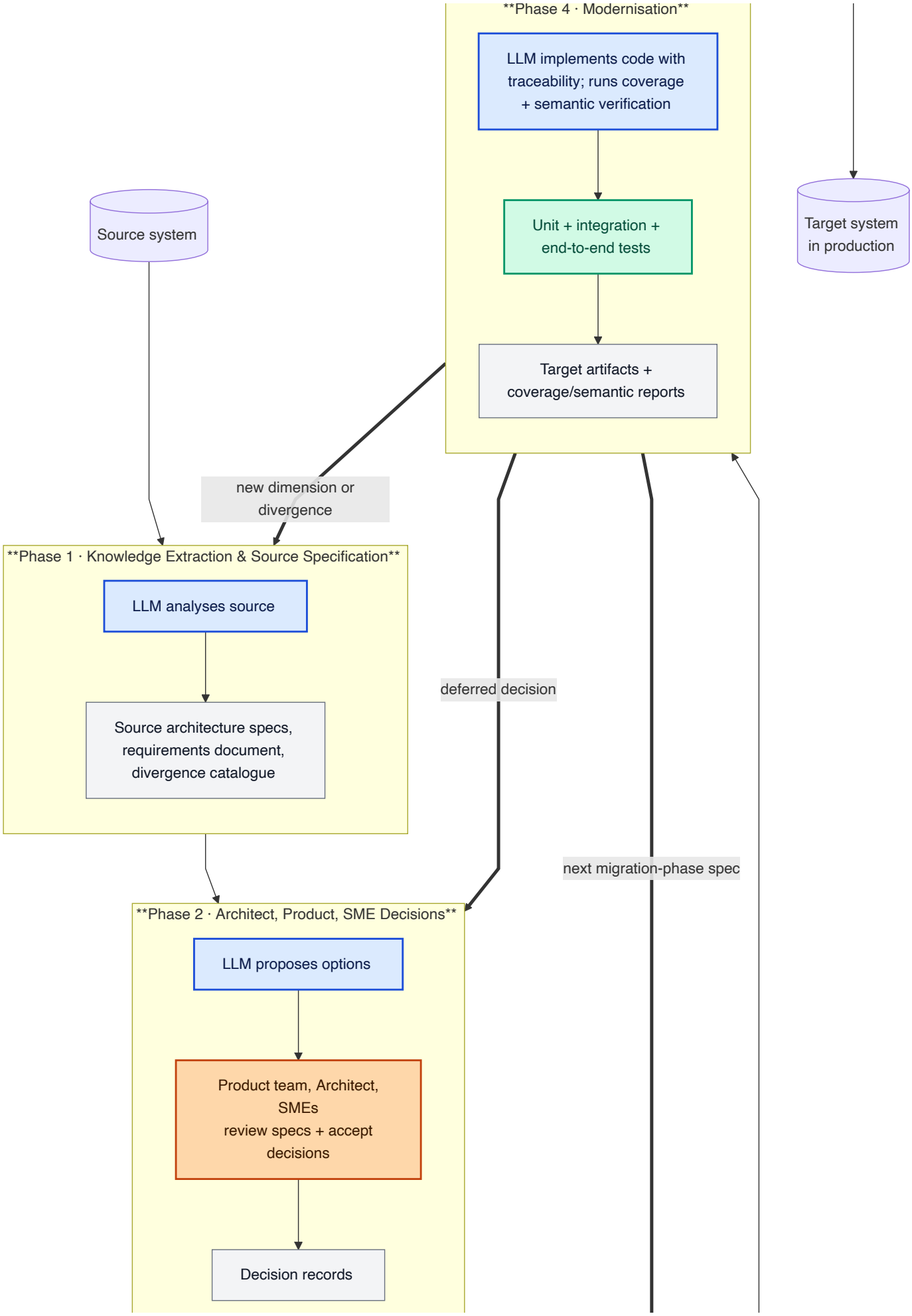
These three work together as an auditable loop. Each addresses a distinct stakeholder need.

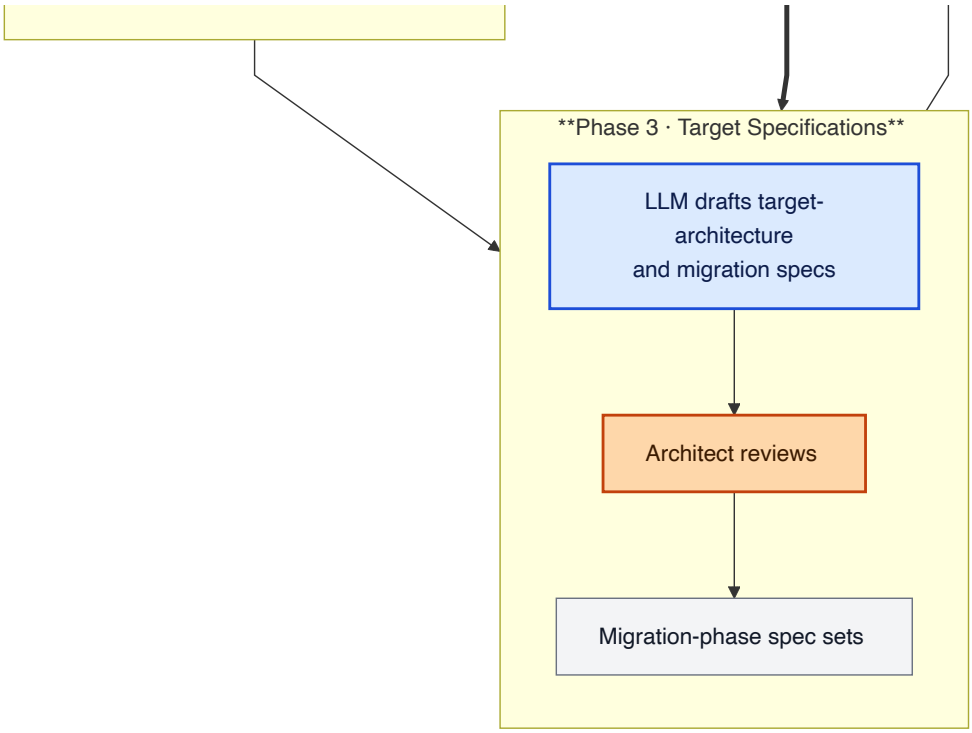
- **② Traceability** — every target element links back to its source origin **and** to the specs and decisions under which it was produced.
  SME benefit: review becomes a mechanical walk of cited pairs — no more hunting for «where did this code come from?»
- **③ Coverage Validation** — every source architecture element is classified as *covered* · *eliminated* · *unmatched*, reported **per dimension**.
  Stakeholder benefit: «done» is a predicate, not a percentage; missing hook sites or unmapped tables are flagged immediately.
- **④ Modernization-Gap Resolution** — unmatched or low-confidence elements trigger either **new** target code or **correction / enhancement** of already-generated target code; new source ↔ target divergences feed the project's divergence catalogue so every later phase benefits.
  Team benefit: confidence to iterate — the target is not frozen after first generation; modernization continues as knowledge deepens.

Slide 5 — The five phases at a glance

For each phase: **who works**, **what they produce**, **how it is validated**.




Phase 4 · Modernisation
LLM implements code with traceability; runs coverage + semantic verification
Unit + integration + end-to-end tests
Target artifacts + coverage/semantic reports
Target system in production
Source system
new dimension or divergence
Phase 1 · Knowledge Extraction & Source Specification
LLM analyses source
Source architecture specs, requirements document, divergence catalogue
deferred decision
next migration-phase spec
Phase 2 · Architect, Product, SME Decisions
LLM proposes options
Product team, Architect, SMEs review specs + accept decisions
Decision records



> **Bold thick arrows** mark mid-phase re-entries; Phase 4 routinely loops back to Phases 1-3, and Phase 5 overlaps Phase 4 during the coexistence period.

Colour legend: blue = LLM work · orange = human judgement · green = automated validation · grey = artifacts produced.

Validation per phase

- Phase 1 output is reviewed by the Product team, Architect and SMEs **inside Phase 2** as part of accepting decisions — that is how knowledge extraction is gated.
- Phases 3 onwards are validated by automated tests (unit + integration + end-to-end) run inside the Phase 4 loop, after every batch of implemented code.

Slide 6 — Phase 1 · Knowledge Extraction & Source Specification

Input: source repository, domain docs, SME recordings, configs, logs.

Output (three artifact families):

- **Source architecture specifications** — one per dimension
- **Requirements document** — Mission / Goals / Premises / Constraints + traceability matrix
- **Platform-divergence catalogue** — behavioural divergences between source and target platforms (living artifact; grows through every later phase)

Reference project example artifacts:

- `specs/architecture/00-project-charter.md` — requirements document
- `specs/architecture/01-architecture.md` .. `15-sme-review.md` — source architecture specs
- `specs/architecture/14-semantic-discrepancies.md` — platform-divergence catalogue

Slide 7 — Phase 2 · Architect, Product, SME Team Decisions

Input: Phase-1 artifacts; research findings.

Output: decision records (context, options, trade-offs, decision, consequences), plus a decision index linking them together.

Who accepts each decision:

Role	Accountable for
Product team	Business impact; scope; keep/deprecate/enhance
Project architect	Target architecture coherence; flow variant
Subject-matter experts	Domain constraints; behavioural parity
Security / Ops / Dev leads	Trust; deployment; implementability

Acceptance modes: up-front · just-in-time · parallel with implementation.

Reference project example artifacts:

- `docs/decisions/README.md` — decision index and dependency graph
- `docs/decisions/0001-migration-flow-variant.md` — a P0 decision (flow variant)
- `docs/decisions/0002-python-framework.md` — a P0 decision (target framework)
- `docs/decisions/0014-feed-parsing-library.md` — a P1 decision (accepted just-in-time)

Slide 8 — Phase 3 · Target Specifications

Each migration phase gets a spec set of three files that describe the same target scope from two angles:

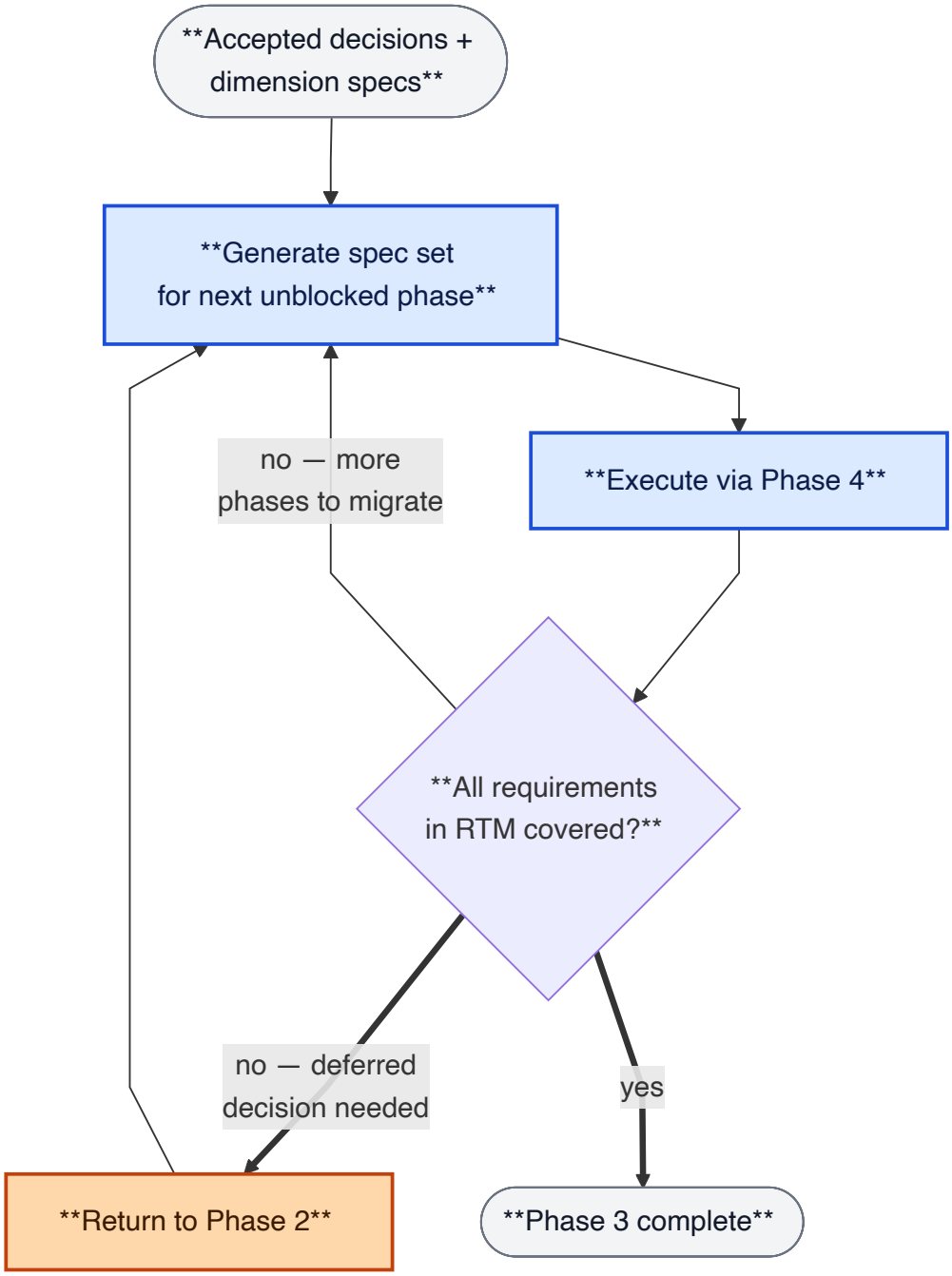
- **Migration-phase spec** — behaviour, functional requirements, acceptance criteria (*«what the target should look like»*)
- **Migration-phase plan** — batches in dependency order, entry and exit gates (*«how we get there»*)
- **Migration-phase tasks** — actionable steps with source cross-references (*«the work to do»*)

Generated **iteratively**: one migration phase at a time; decisions that are still deferred trigger a short loop back to Phase 2.

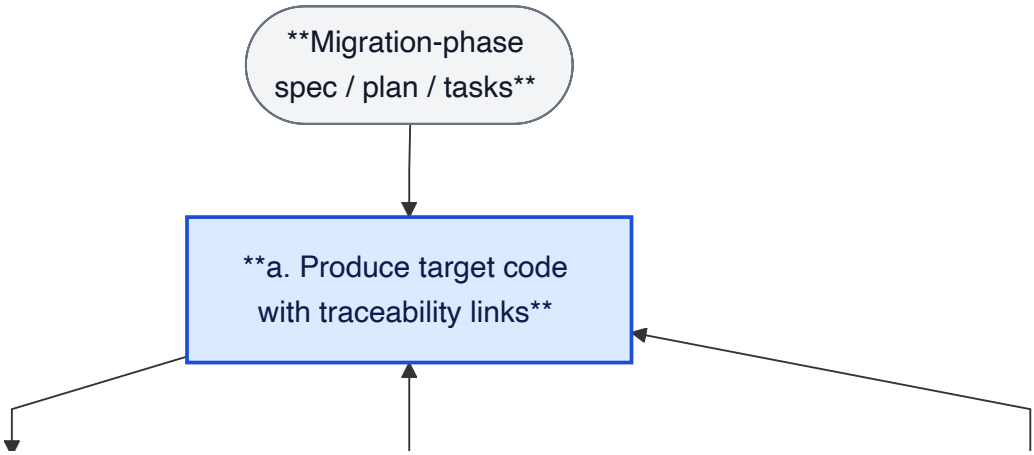
Reference project example artifacts:

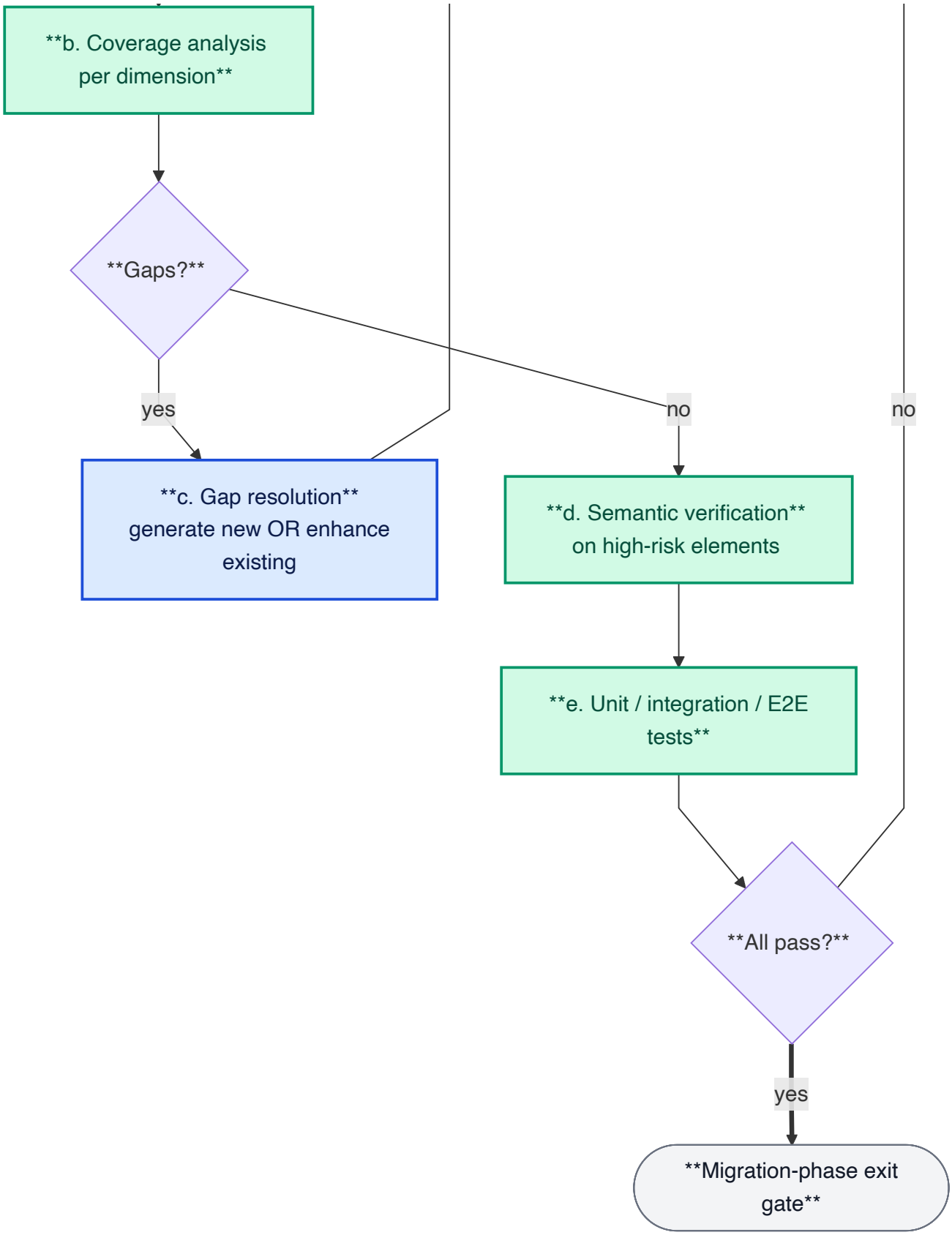
- `specs/001-foundation/{spec,plan,tasks}.md` — foundation (walking skeleton)
- `specs/002-core-logic/{spec,plan,tasks}.md` — core business logic
- `specs/004-api-handlers/{spec,plan,tasks}.md` — API layer
- `specs/005-semantic-verification/{spec,plan,tasks}.md` — verification phase
- `specs/006-deployment/{spec,plan,tasks}.md` — deployment phase

Slide 9 — Phase 3 · Iteration



Slide 10 — Phase 4 · Migration Execution (per migration-phase loop)





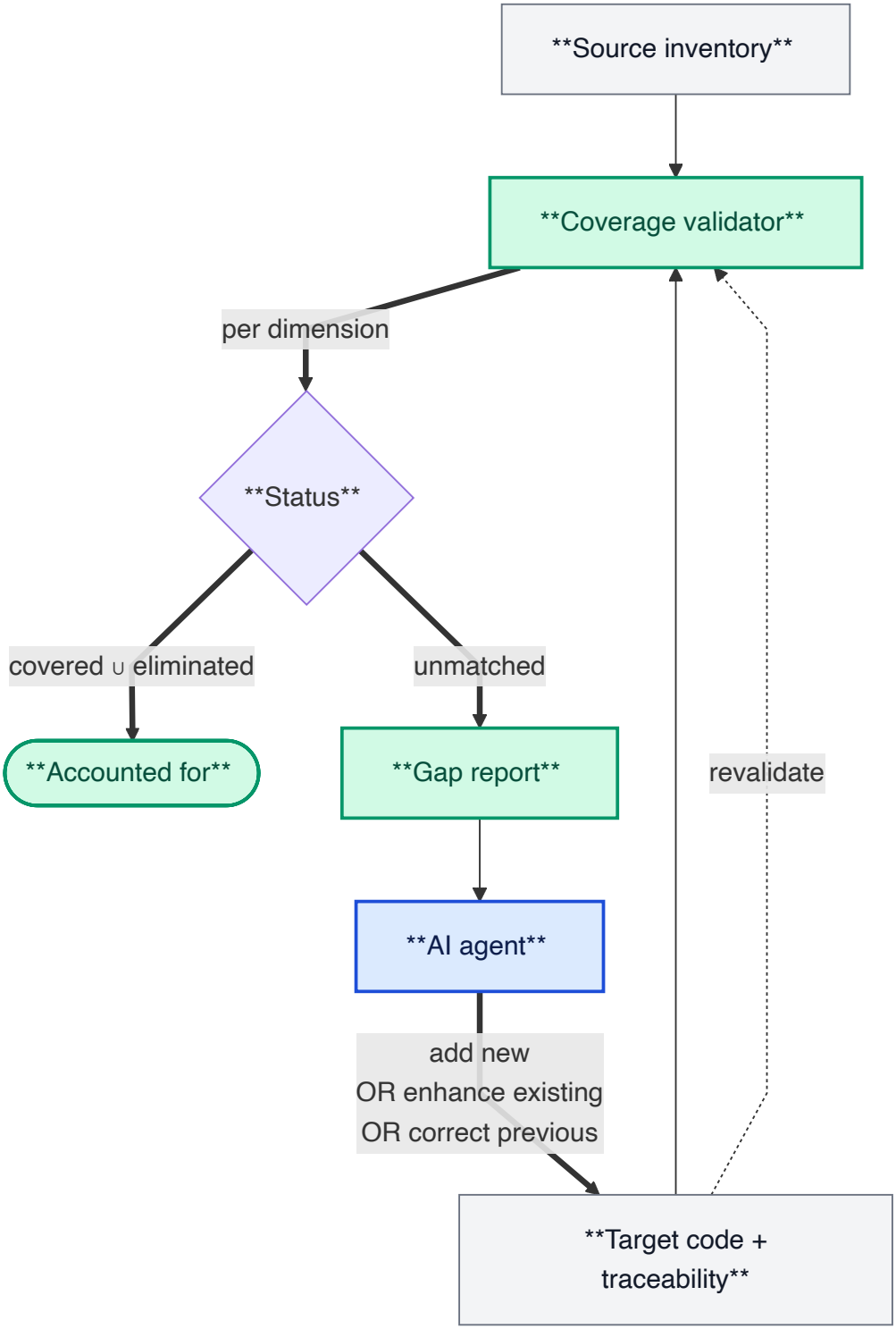
Exit gate: zero unmatched in-scope elements · all tests green · all required decisions accepted.

Reference project example artifacts:

- `target-repos/ttrss-python/` — target code with inline traceability links
- `tools/graph_analysis/validate_coverage.py` — the coverage validator

- `docs/reports/semantic-verification.md` — the semantic verification report

Slide 11 — The core mechanism: traceability, coverage, gap resolution



The agent is **not** told to translate lines. It is told **which source element has not been accounted for**, and decides where it belongs in the target.

Slide 12 — Why generate-or-enhance matters

When the coverage validator flags a source element as unmatched or low-confidence, the agent treats that as a signal to **correct or enhance the existing target element** as well as to add new code. This differs from approaches that can only generate new code in response to a new gap:

Capability	Pure LLM generation	Pure agentic loop	This framework
Can correct already-produced target code	No	In principle	**Yes**
Has a categorical signal for *which* element	No	No	**Yes (per-dimension coverage delta)**
Detects silent behavioural divergence	No	Ad-hoc	**Checklist-driven against catalogue**

Result: gradual **modernisation**, not only line-for-line translation.

Slide 13 — Phase 5 · Hybrid Deployment & Cutover

Goal. Run the target alongside the source under realistic conditions; shift traffic progressively; decommission the source.

Release modes (choose per project):

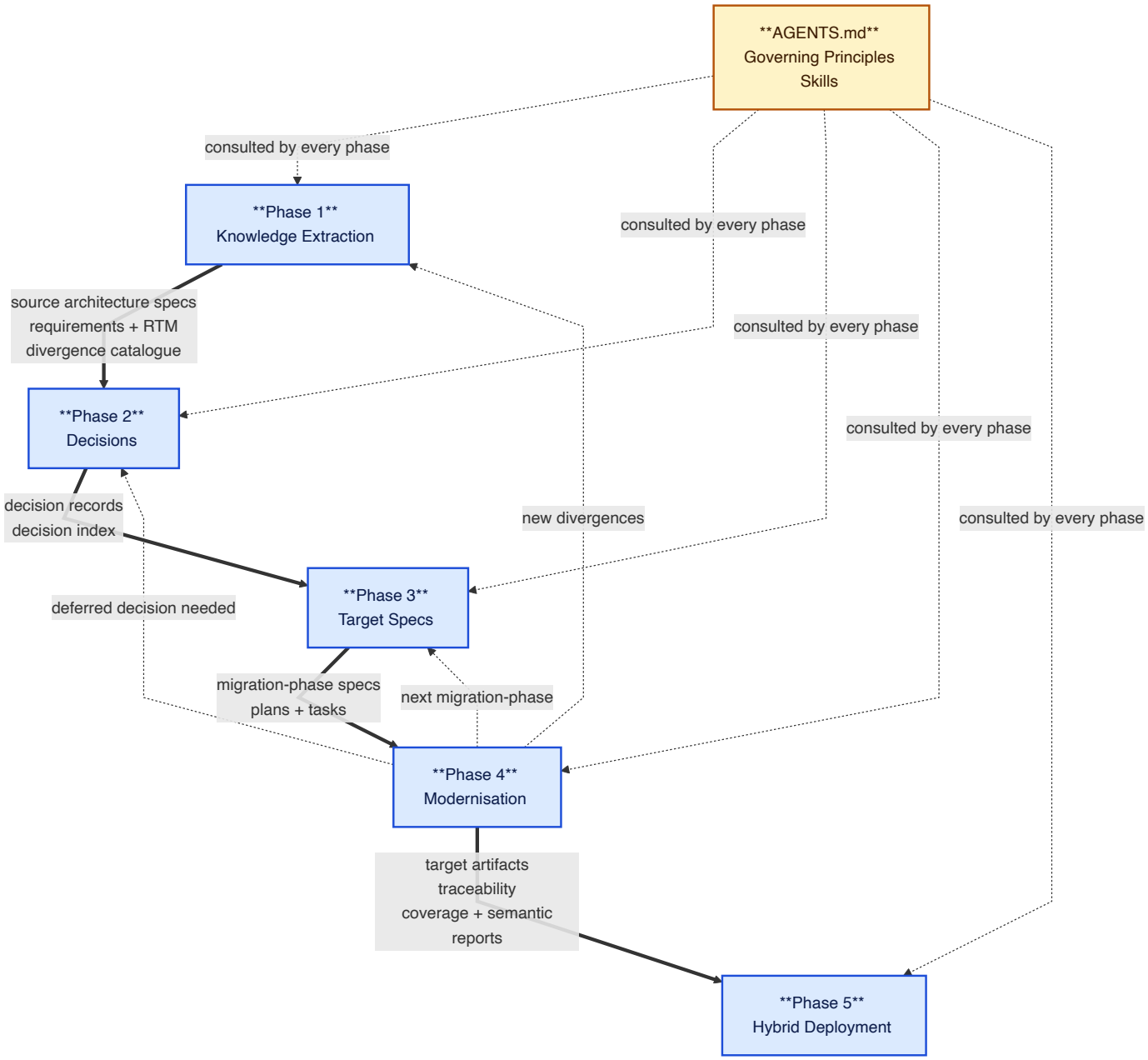
- Big-bang single cutover
- Gradual coexistence
- Incremental traffic shift (gateway / service mesh / feature flag)
- Read-first / write-later
- Parallel-run comparison

Output topology may reshape the repository set — single repo, one-to-many, many-to-one, or many-to-many.

Reference project example artifacts:

- `specs/006-deployment/plan.md` — staging + cutover plan
- `target-repos/ttrss-python/docker-compose.prod.yml` — production compose
- `target-repos/ttrss-python/nginx/nginx.conf` — reverse-proxy config
- `target-repos/ttrss-python/pgloader.load` — data-migration script

Slide 14 — Artifact flow



Slide 15 — Example artifact: a dimension spec

```
# specs/architecture/02-database.md

## Entity / schema dimension

- Tables: 35 (31 active + 4 deprecated)
- Foreign-key graph: ttrss_users → ttrss_feeds → ttrss_entries → ttrss_user_entries
- Migration ordering implied: users first, feeds next, entries last
- Deprecated tables (not ported): legacy themes, v1 labels, v1 filters, ...
```

Source-file cross-references by relative path + line number; no duplication.

Slide 16 — Example artifact: a decision record

```markdown
# 0002 — Select Target Web Framework

## Considered Options

1. Framework A — closest match to source handler dispatch
2. Framework B — modern async, but no built-in sessions
3. Framework C — full-featured but over-engineered

## Decision: Framework A

Rationale: minimises translation distance; preserves client-server contract.

## Consequences

- zero client-side changes

* synchronous runtime; async deferred to workers
```

Every non-trivial choice recorded; accepted by named stakeholders.

Slide 17 — Example artifact: a traceability link

```python
# Source: ttrss/include/functions.php:authenticate_user (lines 706-771)
# Specs:  specs/architecture/06-security.md, specs/001-foundation/spec.md
# Decisions: 0008-password-migration.md
def authenticate_user(login: str, password: str) -> Optional[User]:
    ...
```

Two links:

- **Source link** — what source shaped this code
- **Specs / Decisions link** — which spec and decision justified this shape

---

Slide 18 — Example artifact: a migration-phase spec

```markdown
# specs/001-foundation/spec.md

## Migration Phase 1: Foundation (walking skeleton)

### What the target looks like

- App factory + dependency injection
- Core 10 ORM models (users, feeds, entries, ...)
- Server-side session store

### What must be preserved from the source

- Existing client-server contract unchanged
- Password upgrade path (legacy hash → modern hash on next login)

### How we get there

1. Models (`ttrss/models/`)
2. Auth verifier
3. Login endpoint
```

```
4. Test suite

### Exit gate

- All unit tests green · zero coverage gaps in `auth/` dimension
- ADR-0007, ADR-0008 accepted
```

Slide 19 — Fundamental Benefits

- **Completeness as a predicate.** Every source element is covered, eliminated, or unmatched.
- **Dimensional coverage, not file coverage.** Per-axis reporting, not one number.
- **Generate-or-enhance.** Coverage signals drive correction, not only generation.
- **Unified target-arch and migration-strategy specs.** No drift between two sets of docs.
- **Platform-divergence catalogue.** Ad-hoc review becomes checklist-driven audit.
- **Deferrable decisions.** P1/P2 decisions bound to migration phases, not inception.

Slide 20 — Applicability

System type	Primary relationships that drive migration
Web application	Data — table joins, foreign keys
CLI tool	Code — caller / callee
Network protocol library	State — transitions and timers between states
Microservices	Services — inter-service call and contract edges
Data pipeline	Data flow — source → transform → sink
Event-driven architecture	Events — producer → topic → consumer
Embedded / IoT	Hardware — device ↔ driver ↔ application
Stored-procedure system	Data flow — trigger → procedure → procedure

Same five-phase pipeline; only the dimensions and decisions change.

Slide 21 — When the framework fits

Fits: large codebases, regulated industries, long-running migrations, modernisation (not just translation), projects where «completeness» must be auditable.

Over-kill: small one-off ports (<5,000 LOC), pure code-style refactors, prototypes.

Unique strengths:

- Supports architectural redesign (not only 1:1 translation)
- Provides a per-dimension completeness signal, not a single percentage
- Produces auditable records for every decision and every target element

---

Slide 22 — How to start

1. **Install the governance file** (AGENTS.md) and skill library at the repository root.
2. **Run Phase 1** — deep research on the source + first pass at dimensions.
3. **Run Phase 2** — accept P0 decisions only; defer the rest.
4. **Run Phase 3** for the first unblocked migration phase (often a walking skeleton).
5. **Run Phase 4** — implement, validate, verify, test.
6. **Iterate** — later migration phases trigger loops back to Phase 2 when needed.
7. **Run Phase 5** — hybrid deployment once enough target is ready.

Slide 23 — Summary

> Dimensions make the source system inspectable;
> traceability makes the target auditable;
> coverage and gap-resolution together make **completeness** a measurable, achievable property —
> which is what turns AI-assisted migration into something you can actually ship.